The 4% Rule Under Return Clustering

Does it matter that stock returns come in runs? An independent research project

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Abstract

Suppose a retiree puts $1,000,000 entirely in U.S. stocks, withdraws 4% ($40,000) in the first year, and takes out the same amount in inflation-adjusted terms every year for 30 years. In U.S. history, that retiree would have run out of money in only 2 of 113 retirements (98.23% success). When I simulate returns as independent random draws, success drops to about 90%. Block bootstrap simulations, which resample runs of consecutive years, close part of that gap: from 90.30% with 3-year runs to 96.15% with 20-year runs. That fits the idea that returns cluster and that clustering matters, but it doesn't prove it. Long runs start to replay history itself, and the 113 historical retirements overlap so much that they amount to only about four or five independent periods.

1. Introduction

The 4% rule says a retiree can withdraw 4% of the starting portfolio in year one, raise that dollar amount with inflation each year, and expect the money to last 30 years. It comes from backtests, which replay actual U.S. returns (Bengen, 1994; Cooley, Hubbard and Walz, 1998). A backtest runs each real 30-year stretch exactly as it happened, so the order of returns, including runs of good or bad years, stays intact.

Simulations, by contrast, often draw each year's return independently. That wipes out any clustering: a bad year is no more likely to follow a bad year than a good one. The question is whether that matters. It could, because bad returns early in retirement do the most damage. They shrink the balance that every later withdrawal comes out of.

I compare the historical success rate with three kinds of simulation. The first draws each year independently from a normal distribution (the bell curve). The second is the i.i.d. bootstrap. A bootstrap resamples the actual historical returns instead of assuming a formula for them, and “i.i.d.” means independent and identically distributed: every year is an unrelated draw from the same pool. The third is the block bootstrap, which resamples runs of 3, 5, 10, 15 or 20 consecutive years and keeps each run in its original order. If clustering matters, success rates should move toward the historical result as the runs get longer.

2. Data

The data come from Robert Shiller's annual U.S. series (Chapter 26 of *Market Volatility*, MIT Press, 1989, updated), downloaded as `chapt26.xlsx` from his data page (econ.yale.edu/~shiller/data.htm). I use the “Return on S&P Composite” column: 142 years, 1871 to 2012. Real prices run to 2015, but the real dividend series stops in 2012, and a return needs both, so 2012 is the last year. “Real” means adjusted for inflation. The mean real return is 0.0804 (8.04% a year). The standard deviation, which measures how much returns swing from year to year, is 0.1772 (17.72%).

The return column is already adjusted for inflation, so I use it as it is. Adjusting it a second time with CPI (the consumer price index) would count inflation twice and understate returns. I checked this instead of

assuming it. From the real price and real dividend series, I recomputed each year's return as (next year's real price + this year's real dividend) / this year's real price - 1. The biggest difference from the column, across all years, is about 2e-16. That is computer rounding error, so the two agree.

3. Method

The retiree starts with B_0 = \$1,000,000, all in stocks, and withdraws w = \$40,000 (4% of the starting balance) every year. Because the returns are already adjusted for inflation, a constant w is the same as a withdrawal that rises with inflation. Each year, the return is applied first and the withdrawal comes second:

$$B_t = B_{t-1} (1 + r_t) - w$$

A path is one simulated retirement. It fails in the first year the balance reaches zero, and it stays at zero after that. If the money lasts all 30 years, it succeeds.

Four methods generate the 30 annual returns:

1 **Historical windows.** Every consecutive 30-year stretch in the data. With 142 years, there are 142 - 30 + 1 = 113 windows, starting in 1871 through 1983.

2 **i.i.d. normal.** Each year is drawn independently from a normal distribution with the same mean and standard deviation as the data.

3 **i.i.d. bootstrap.** Each year is drawn independently, with replacement, from the 142 historical returns. “With replacement” means a year can be picked more than once.

4 **Block bootstrap.** Pick a random start year, copy the next *b* years in their original order, and repeat until you have 30 years (the last block is cut short if needed). I use *b* = 3, 5, 10, 15 and 20. Blocks don't wrap around the end of the data, so a block of length *b* can start in any of 142 - *b* + 1 positions.

Each simulated method runs 10,000 paths with a fixed seed of 42 (NumPy's default random number generator), so the numbers can be reproduced exactly. I checked the code with 12 automated tests, including a hand-computed example: returns of +10%, -50% and 0% take a \$1,000,000 portfolio to \$1,060,000, then \$490,000, then \$450,000 (balances after each year's withdrawal).

4. Results

History succeeds in 98.23% of windows, every independent-draw simulation lands near 90%, and block bootstraps climb from 90.30% to 96.15% as blocks lengthen.

Method	Paths	Success rate	95% interval (pp)	Failed paths
Historical windows	113	98.23%	±2.4	2
i.i.d. normal	10,000	89.92%	±0.6	1,008
i.i.d. bootstrap	10,000	89.71%	±0.6	1,029
Block bootstrap, 3 years	10,000	90.30%	±0.6	970
Block bootstrap, 5 years	10,000	90.59%	±0.6	941
Block bootstrap, 10 years	10,000	91.59%	±0.5	841
Block bootstrap, 15 years	10,000	92.91%	±0.5	709
Block bootstrap, 20 years	10,000	96.15%	±0.4	385

The 95% intervals use 1.96 × sqrt(p(1 - p)/n), where p is the success rate and n is the number of paths. For the historical row, this treats the 113 windows as independent. They are not, so the true uncertainty is larger.

Two historical cohorts fail, meaning the retirees who started in 1929 and in 1966. Their money ran out in retirement years 27 and 28, late in the 30 years. Many simulated failures fall in that range too (Figure 3).

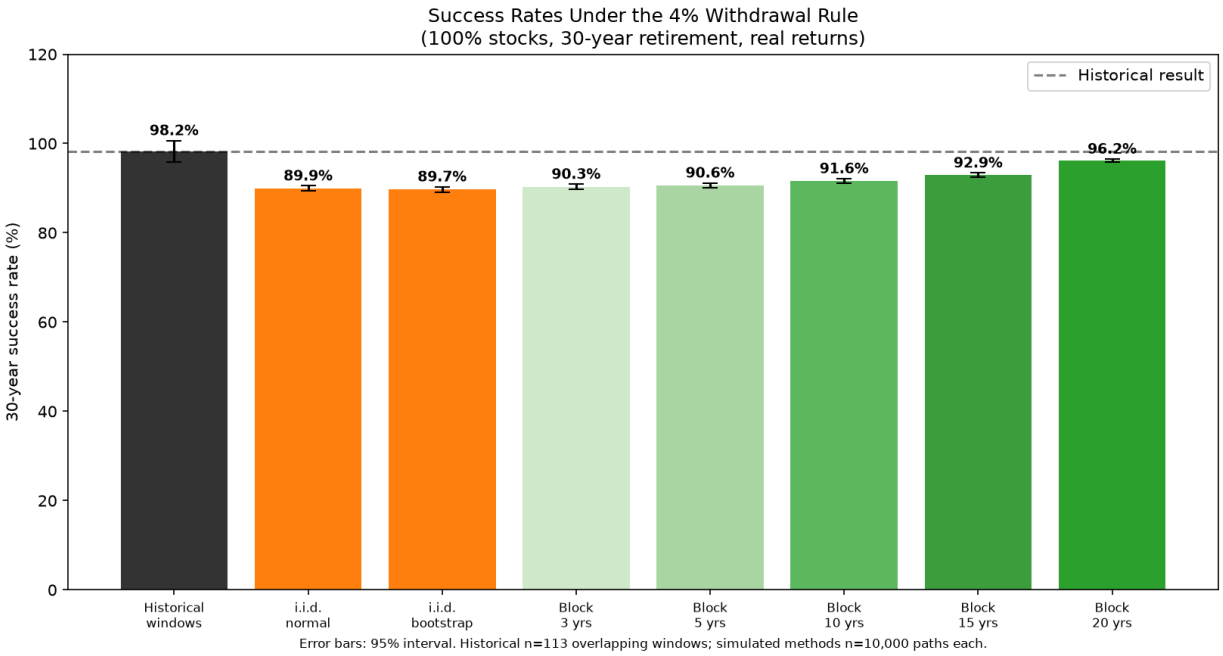


Figure 1. Success rate by method, with 95% intervals. The dashed line is the historical result.

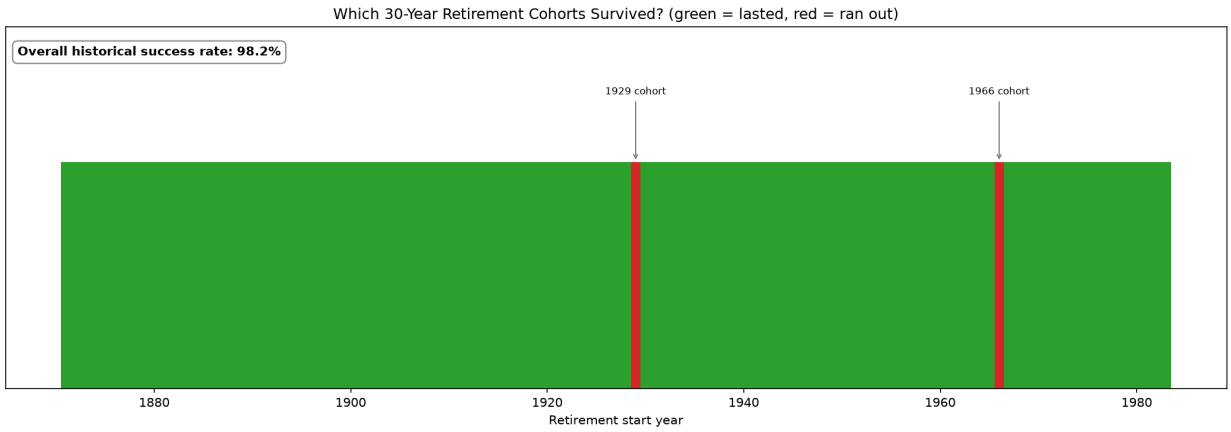


Figure 2. Outcome of each historical 30-year retirement by start year. Red marks the 1929 and 1966 cohorts.

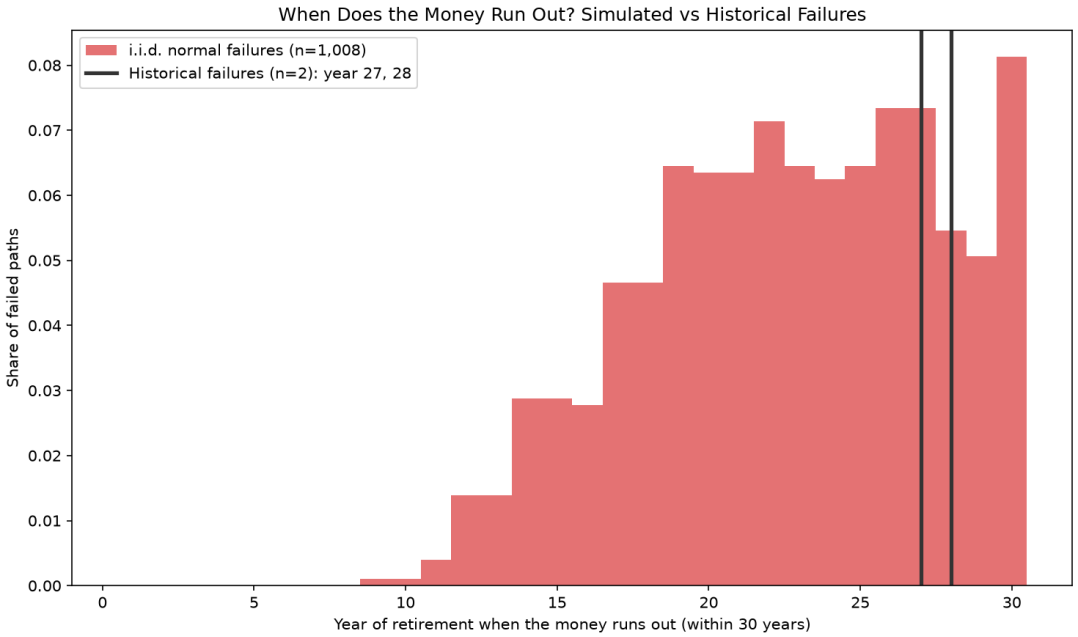


Figure 3. Year of retirement in which the money runs out: simulated i.i.d. normal failures (n = 1,008) against the two historical failures.

5. Discussion and limitations

The simulations are more pessimistic than history, and the gap shrinks as the blocks get longer. The i.i.d. normal simulation uses the data's own mean and standard deviation, yet it succeeds 8.3 percentage points less often than history (89.92% against 98.23%). So average return and volatility (the size of the swings) don't explain the historical result on their own. Compared with the i.i.d. bootstrap (89.71%), the block bootstraps gain 0.59, 0.88, 1.88, 3.20 and 6.44 points at 3, 5, 10, 15 and 20 years. With 10,000 paths each, two simulated rates can differ by about 0.8 points from chance alone, so only from 10-year blocks upward is the gain clearly larger than noise. The two i.i.d. methods differ by just 0.21 points, so swapping the bell curve for the actual returns changes almost nothing. In these tests, what moves the result is keeping runs of years together.

That fits the idea that clustering matters, but it doesn't prove it. There are four reasons.

- **Long blocks replay history.** A 30-year path built from 20-year blocks is mostly two stretches of real history stuck together. So some of the move toward the historical rate is automatic.
- **History is a small sample.** The 113 windows overlap heavily. 142 years holds only about four to five separate 30-year periods, so the 98.23% is much less certain than it looks.
- **No direct test of dependence.** I compare success rates across resampling methods. I don't test the returns directly for autocorrelation (whether a return is linked to the one before it) or mean reversion (whether unusually bad years tend to be followed by better ones).
- **Narrow scope.** The data cover one country, 1871 to 2012, at yearly frequency. The retiree holds 100% stocks, withdraws a fixed real amount with no flexibility, and pays no fees or taxes. Blocks don't wrap around the end of the data, and I picked the block lengths in advance instead of using a rule to choose them.

Next steps could include autocorrelation and variance-ratio tests on the series (the second checks whether the variance of multi-year returns grows in step with the horizon, as it would if years were independent), a stationary bootstrap that uses random block lengths, other horizons and withdrawal rates (all set in one block at the top of `run.py`), and data from other countries.

6. Conclusion and reproducibility

The 4% rule looks safer in the historical record (98.23%) than under independent random returns (about 90%). Keeping runs of consecutive years in the simulation closes much of that gap as the runs get longer. That makes clustering a plausible explanation, but not a demonstrated one.

Every number here comes from the project code. It loads Shiller's workbook, checks the return column, runs all the methods with a fixed seed, and draws the figures. The 12 automated tests all pass. They cover the withdrawal rule, how failures are recorded, the window counts, whether the bootstrap gives the same results with the same seed, and the dataset's known summary numbers. The code and figures are in the project's GitHub repository: `https://github.com/zshqv/Retirement-Monte-Carlo-Simulator`. I don't redistribute Shiller's data. Download it from his page and rebuild it with the included script.

References

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